

monitoring the chamber (when current is reduced by a predetermined amount, a threshold is crossed and an alarm condition is established). A dual-ionisation chamber is required to compensate for the possible effects of humidity and pressure changes (changes in humidity and/or atmospheric pressure affect chamber current and create an effect similar to the effect of particles of combustion entering the sensing chamber).

A dual-chamber detector utilises two ionisation chambers. One is a sensing chamber, affected by particulate matter, humidity and atmospheric pressure, which is open to outside air. The other is a reference chamber, which is partially closed

to outside air and is affected only by humidity and atmospheric pressure, because its tiny openings block the entry of larger particulate matter including particles of combustion.

An electronic circuitry monitors both chambers and compares their output. If humidity or atmospheric pressure changes, outputs of both chambers are affected equally and cancel each other out. When combustion particles enter the sensing chamber, its current decreases while the current of the reference chamber remains unchanged. The resulting current imbalance is detected by the built-in electronic circuitry.

Characteristics of an ionisation detector make it more suitable for detection of fast-flaming fires that are characterised by combustion particles in the 0.01 to 0.4 micron-size range.

Beware! It is illegal to disassemble an ionisation smoke detector and remove the tiny radioactive source.

Also, with the obvious potential for the spread of radioactive contamination, once the Americium source is removed from its holder, it becomes extremely difficult to arrange safe disposal.

Advanced smoke detectors

Smoke detectors offer the earliest possible warnings of fire. In cer-

tain circumstances where standard smoke detectors are unsuitable, special-purpose detectors may be used. For example, laser based smoke detectors are designed for use in areas that require extremely early warnings of fire. These are designed to detect the earliest particles of combustion, making these ideal for any area with mission-critical operations.

Laser based smoke detectors are ultra-sensitive to smoke, being as much as 100 times more sensitive than standard ones.

Aspiration smoke detectors use a pipe and fan system to draw smoke particulates back to the detection chamber. Most air-sampling detectors are aspiration smoke detectors that work by actively drawing air through a network of small-bore pipes laid out above or below a ceiling in parallel, covering a protected area. Small holes drilled into each pipe form a matrix of holes (sampling points), providing an even distribution across the pipe network. Air samples are drawn past a sensitive optical device, often a solid-state laser, tuned to detect extremely small particles of combustion.

Air-sampling detectors may be used to trigger an automatic fire response. Likewise, some smoke detectors use a carbon-dioxide sensor or carbon-monoxide sensor in order to detect extremely-dangerous products of combustion.

Lately, designers have begun to introduce smoke detectors that combine photoelectric and ionisation smoke sensors in the same unit. These units combine the advantages of both the sensors into an advanced unit that detects smoke from a broader spectrum of fires. A residential fire-detection system may also include a heat detector.

There are two types of heat detectors: fixed-temperature and rate-of-rise. Fixed-temperature detectors are configured to wake up when

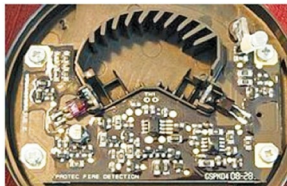


Fig. 3: Inside view of a light-scattering smoke detector



Fig. 4: Radioactive button used in most ionisation smoke detectors

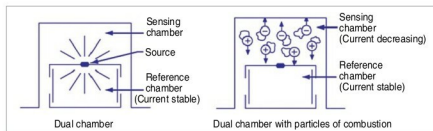


Fig. 5: Dual chamber in an ionisation smoke detector